Tracy (Huiwen) Han

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Education

Boston University *Boston, MA*
B.A. in Biology; minors in Computer Science and Business Administration; GPA 3.7/4.0 *2020 – 2023*

Research Experience

Brain Stimulation Lab, Stanford University *Stanford, CA*
Research Assistant, working on Precision Neuromodulation *Sep. 2024 – Present*
Advised by Christopher Austelle and David Spiegel.

Physical Automation Intelligence Lab, Texas A&M University *College Station, TX*
Visiting Student, working on 3D Generative Models and Agentic RL *Jun. 2025 – Present*
Advised by Zhiwen Fan.

Research Interests

3D generative models; long-horizon reinforcement learning; continual learning.

Publications

[1] **Huiwen Han**[*†], Lulin Liu[*], Bangya Liu, Yuanhao Cai, Nuo Chen, Xiaoqing Wang, Ziqian Xie, Chenyu You, Shuiwang Ji, Degui Zhi, and Zhiwen Fan. "AnaDiffusion: Anatomically Compositional Latent Diffusion for Controllable 3D Brain MRI Generation." In *Advances in Neural Information Processing Systems* (**NeurIPS**), 2026. arXiv:2608.23014.

[2] **Huiwen Han**[*†], Ziyao Zeng, Zihao Zhu, Sisong Bei, Hao Wang, Hongyu Xu, Yan Han, Degui Zhi, and Zhiwen Fan. "Contextual Credit Policy Optimization," under review, 2026. arXiv:2610.19103.

[3] **Huiwen Han**[*†], Adam R. Pines, Yixin Wang, Yu Zhang, and Christopher W. Austelle. "NetHGL: Hierarchical Graph Learning Identifies Remission-Associated Network Organization in Treatment-Resistant Depression Following Stanford Neuromodulation Therapy."

[4] **Huiwen Han**[*†], Nuo Chen[*], Ziyao Zeng[*], Dayou Li, Hao Wang, Amir Hossein Rahmati, Xiaoning Qian, Bangya Liu, and Zhiwen Fan. "TouchTwin: Scaling Human Touch Data via Real-to-Sim-to-Real." In *Conference on Robot Learning* (**CoRL**), Late Breaking Results, 2026.

[5] Minseong Kweon, Wenyuan Zhao, Nuo Chen, Lulin Liu, **Huiwen Han**, Zihao Zhu, Srinivas Shakkottai, Chao Tian, and Zhiwen Fan. "DarkVGGT: Seeing Through Darkness Using Thermal Geometry without Daylight Tax." In *Advances in Neural Information Processing Systems* (**NeurIPS**), 2026. arXiv:2606.11326.

[6] Tian Xia[*], Nuo Chen[*], Zihao Zhu, **Huiwen Han**, Ziqian Xie, Zhiwen Fan, and Degui Zhi. "Surface-volume self-supervised representation learning of brain MRI for genetic discovery." arXiv:2610.02114, 2026.

[7] Kenneth Shinozuka, Clayton Olash, **Huiwen Han**, Azeezat Azeez, Malvika Sridhar, Andrew D. Geoly, Catherine Daye, Saron Hunegnaw, Kirsten N. Cherian, Jackob N. Keynan, Randi E. Brown, Derrick M. Buchanan, John P. Coetzee, Ian H. Kratter, Raag D. Airan, Martijn Arns, Maheen M. Adamson, Manish Saggar, and Camarin Rolle. "Dynamic Substates of the Default Mode Network Are Associated with Mystical Experiences and Clinical Outcomes after Magnesium-Ibogaine Treatment in Veterans with Traumatic Brain Injury." medRxiv, 2026. doi:10.64898/2026.07.20.26358512.

[8] Florence X. Doo, **Huiwen Han**, Bradley A. Maron, and Heng Huang. "Trustworthy Clinical Thinking in MLLMs: Hierarchical Energy-Based Reasoning for Interpretable Medical Scans (HERMES)." In *AI for Clinical Applications*, LNCS, Springer, 2025. Accepted for an **oral presentation** at the 1st Workshop on Multimodal Large Language Models in Clinical Practice, 28th International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2025.

Research Service

Reviewer for *Transactions on Machine Learning Research*.
Reviewer for *IEEE Transactions on Circuits and Systems for Video Technology*.
Co-reviewer for *Psychiatry and Clinical Neurosciences*.
Co-reviewer for *CNS Neuroscience & Therapeutics*.

Projects

- **TouchTwin** (CoRL 2026 Late Breaking Results). A real-to-sim-to-real pipeline that scales human touch data: rebuilds tactile-glove recordings in MuJoCo with coding-agent-written, physics-tested objects; uses recorded touch to drive physics replay, recovering contact force that vision misses; and expands replays into perturbed demos and simulation-conditioned video, yielding hand policies with human-level grip.
- **AnaDiffusion** (NeurIPS 2026; arXiv). Anatomically compositional latent diffusion for controllable 3D brain MRI generation; improves anatomical realism when generating 3D brains across Alzheimer's disease, mild cognitive impairment, and healthy aging.
- **Contextual Credit Policy Optimization (CCPO)** (under review, 2026; arXiv). A critic-free reinforcement learning method for long-horizon LLM agents that assigns dual contextual step-level credit from historical returns and future progress; future credit stabilizes learning as the historical signal diminishes over training, and CCPO substantially reduces repetitive agent behaviors for efficient multi-step task execution.
- **NetHGL**. Hierarchical graph learning that models remission-associated network organization in treatment-resistant depression following Stanford Neuromodulation Therapy, leveraging structural priors to represent the brain as a dynamic system; identifies a candidate treatment-response biomarker.
- **DarkVGGT** (NeurIPS 2026; arXiv). An RGB-T feed-forward 3D geometry framework that uses physics-aware thermal modeling (thermal factorization and geometry-shared thermal routing) for robust depth and camera pose estimation in dark, low-visibility scenes, while largely preserving performance in well-lit environments.
- **MEVA (Mesh-Enhanced Volumetric Autoencoder)** (arXiv 2026). A self-supervised framework for genetic discovery that encodes voxel-level brain MRI intensity together with cortical mesh geometry (curvature and cortical thickness) into shared imaging features; used as GWAS phenotypes in the UK Biobank, they reveal more genome-wide significant loci than volume-only or mesh-only features.
- **Dynamic DMN substates after magnesium-ibogaine** (medRxiv 2026). Hidden Markov models of resting-state fMRI from an open-label trial of magnesium-ibogaine in 30 U.S. veterans with traumatic brain injury; a dynamic default mode network substate, rather than static connectivity, changed after treatment and correlated with mystical-experience scores and one-month clinical improvement.
- **Multimodal clinical-trial data infrastructure**. Automates reproducible, auditable workflows for high-risk clinical data; supports tiered access control and lays the groundwork for an agentic research system. Funded by a $20,000 Stanford CTS Pilot Grant (2026).
- **Evidence-grounded agents**. A lunar olivine science agent with the Planetary Science Institute, and a multimodal agent that uses video for longitudinal behavioral profiling and symptom monitoring in children with autism spectrum disorder (ASD).

Awards

Stanford Clinical & Translational Science (CTS) Pilot Grant ($20,000) *2026*
Awarded through the Stanford Program on Research Rigor and Reproducibility (SPORR), which funds projects that enhance the validity of research processes and findings, for a multimodal clinical-trial data infrastructure that automates reproducible, auditable workflows for high-risk clinical data. Conceived the idea, drafted the proposal, and served as Lead Scientist and lead engineer/system architect.

* First or co-first author. † Project lead.